Number 4 — Compound percentages · Inverse percentages

Calculator allowed. Give money to the nearest cent and other answers to 3 s.f. Marks in brackets. ★ = harder question.

1 **a** Increase €1500 by 12%. [4]

b Decrease €1500 by 12%.

> **a** Math input error. **b** Math input error.

2 A painting is bought for €40 000. It gains 8% in value in the first year and a further 5% in the second year. Find its value after the two increases. [3]

> Multiply by both factors: Math input error. (Not $+13\%$: the 5% is taken on the increased value.)

3 A flat is valued at €800 000. It loses 15% of its value one year and a further 3% the next. Find its value after the two decreases. [3]

> Math input error.

4 A footballer is valued at €24 million. After a bad season his value depreciates by 20%; after the next, good, season it appreciates by 20%. Find his value after the good season. Is he back to €24 million? [3]

> Math input error million. No — the 20% rise is on a smaller amount than the 20% fall, so he ends 4% below the start ($0.8 \times 1.2 = 0.96$).

5 A diamond is sold for \$38 million. If it appreciates at 7% p.a. for the next 5 years, what is its value after 5 years? [4]

> Multiplier 1.07 applied 5 times: $38 \times 1.07^5 = 38 \times 1.4026 = \$53.3$ million (3 s.f.).

6 A cheetah's top speed is 110 km/h. Suppose it loses 7% of its top speed each year. Find its top speed after 5 years. [4]

> $110 \times 0.93^5 = 110 \times 0.6957 = 76.5$ km/h (3 s.f.).

7 **a** A sun-hat is sold at 25% off for €18. Find its original price. [4]

b A drone is sold for €2100, making a profit of 25%. Find the price the seller paid for it.

> **a** Original p: Math input error.
>
> **b** Math input error. (Not 2100×0.75 — the 25% is of the cost price, not the selling price.)

8 A sculpture is valued at €x. It appreciates by 2% p.a. for 2 years, then depreciates by 3% p.a. for 3 years, and is then sold for €4 million. Find x, to the nearest €1000. [4]

> $x \times 1.02^2 \times 0.97^3 = 4\,000\,000$. The overall multiplier is $1.0404 \times 0.912673 = 0.94954$, so Math input error (nearest €1000).

Total: 29 marks

Check: n years at $R\%$ → multiply by $\left(1 \pm \frac{R}{100}\right)^n$ · successive changes multiply · to find an *original* amount, divide by the multiplier · a 20% fall then a 20% rise is not a return to the start.